MIL-M-43241A

29 December 1971
SUPERSEDING
MIL-M-43241
29 June 1964

MILITARY SPECIFICATION

MILK, FILLED, DRY, PLAIN OR CHOCOLATE, FORTIFIED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers fortified, dry (powdered), filled milk, plain and chocolate, for limited use by the Armed Forces, when authorized, as a nonperishable item (see 6.1).

1.2 Classification.- The product shall be of the following types, as specified (see 6.2):

Type I - Milk, filled, fortified, plain, dry (powdered).
Type II - Milk, filled, fortified, chocolate, dry (powdered).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
JJJ-C-501 - Cocoa.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-F-35081 - Fat, Hydrogenated, Vegetable or Animal.

FSC 8910

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-671 - Sanitary Standards for Milk Evaporating and Drying Plants.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents, form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issues in effect on date of invitation for bids or request for proposal shall apply:

U. S Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Directory of Consumer Protection Programs, Circuits and Officials

Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

Bacteriological Analytical Manual

(Application for copies should be addressed to the Food and Drug Administration, U.S. Department of Health, Education and Welfare, Washington, D.C. 20402.)

U. S. Department of Agriculture

U.S. Standards for Grades of Nonfat Dry Milk (Spray Process)

General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service

Dairy Plants Surveyed and Approved for USDA Grading Service

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(Application for copies should be addressed to the U.S. Department of Agriculture, Dairy Division Consumer and Marketing Service, Washington, D.C. 20250.)

Brucellosis Eradication - Recommended Uniform Methods and Rules, Agricultural Research Service, U. S. Department of Agriculture, Bulletin ARS 91-79.

(Copies may be obtained from U. S. Department of Agriculture, Agricultural Research Service, Washington, D. C. 20250.)

Association of Vitamin Chemists

Methods of Vitamin Assay

(Application for copies should be addressed to John Wiley and Sons, Inc., 605 Third Avenue, New York, N. Y. 10016.)

The United States Pharmacopoeial Convention, Inc.

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 46 Park Ave., New York, N.Y. 10016.)

American Dry Milk Institute, Inc.

Standards for Grades of Dry Milks Including Methods of Analysis; Bulletin No. 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 N. Franklin St., Chicago, Ill. 60606.)

American Public Health Association

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1015 18th Street, N. W., Washington, D. C. 20036.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

* 3. REQUIREMENTS

3.1 Material.-

* 3.1.1 Milk.- The raw milk used in the preparation of this product shall comply with all the quality requirements for No. 1 specified in the General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service. The milk shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the U.S. Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in:

(1) A Modified Accredited Tuberculosis Area; and
(2) Either (a) A certified Brucellosis-Free Area; or
(b) Modified Certified Brucellosis Area; or
(3) An area in the process of being accredited or certified by the USDA in accordance with USDA Agricultural Research Service Bulletin 91-79.

The milk shall be practically free from colostrum, shall be fresh, wholesome, and normal in appearance and odor and shall be subject to inspection by the procuring agency or duly authorized representative.

* 3.1.1.1 Nonfat milk solids.- Fluid skimmed milk and concentrated skimmed milk shall be plant separated from milk complying with 3.1.1. Fluid milk products shall possess a fresh, sweet and pleasing odor and an appearance normal for the product. Nonfat dry milk shall comply with the requirements for U. S. extra grade, low heat product in the Standards for Grades of Nonfat Dry Milk (Spray Process) and shall have been produced in a plant listed by the U. S. Department of Agriculture in the Document titled "Dairy Plants Surveyed and Approved for USDA Grading Service" and approved for the product. Nonfat dry milk shall be not more than 60 days old at time of use.

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3.1.2 Fat.- Fat shall be refined, bleached, deodorized, hydrogenated vegetable fat conforming to the commodity requirements for style A, product of MIL-F-35081.

3.1.3 Lactose.- Lactose shall be of food grade.

3.1.4 Salt.- Salt shall be free-flowing, evaporated, sodium chloride of food grade quantity. Iodized salt shall not be used.

3.1.5 Cocoa.- Cocoa shall comply with type II, style B requirements of JJJ-C-501.

3.1.6 Vanilla flavoring.- Vanilla flavoring, if used in preparation of type II product, shall consist of ground vanilla beans (vanilla powder), vanilla extract, imitation vanilla extract, synthetic vanillin or ethyl vanillin, or combinations thereof. When the vanilla flavoring is in the dry form, it may contain carrying agents such as lactose, dextrose, starch, tricalcium phosphate, or other food grade carriers approved by the Federal Food and Drug Administration.

3.1.7 Stabilizing salts or dispersing agent.- Stabilizing salts or dispersing agents shall be of food grade quality and may be disodium phosphate, tricalcium phosphate, sodium citrate, combinations thereof, or others which are suitable and have been authorized for use in foods under the Food Additives Amendment of the Federal Food, Drug, and Cosmetic Act. If used, the quantity thereof shall not be more than 0.3 percent of the weight of the finished product.

3.1.8 Emulsifying agents.- Emulsifying agents shall be food grade lecithin, mono or di-glycerides or other suitable emulsifiers which have been authorized for use in foods under the Food Additives Amendment of the Food, Drug, and Cosmetic Act. If used, the quantity of such agents shall be limited to not more than 0.5 percent of the weight of the finished product.

3.1.9 Sugar.- Sugar shall be white, refined, cane or beet sugar, or a combination thereof, of food grade quality.

3.1.10 Vitamins A and D.- Vitamins A and D shall be any appropriate type approved by the Federal Food and Drug Administration for use in the fortification of foods.

3.1.11 Antioxidants.- Antioxidants shall be those which have been authorized for use in foods under the Food Additives Amendment of the Federal Food, Drug, and Cosmetic Act. The amount of such agents shall be limited to not more than 0.02 percent by weight of the finished product.

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3.1.12 Coloring.- Coloring, if used in type I product, shall be pure beta carotene, or other appropriate type approved for use in foods by the Federal Food and Drug Administration.

3.2 Processing.- Storage of milk upon receipt at the processing plant, pasteurization of milk, and subsequent processing shall be as defined in the publication referenced in 3.1.1.

3.3 Preparation.-

* 3.3.1 Storage of fluid milk and milk products.- All fluid milk products unless processed within two hours following receipt shall be cooled to 45°F. or lower but not frozen and held in this temperature range until processed.

* 3.3.2 Preparing the blend.- The product shall be prepared from the ingredients in 3.1 to produce a product meeting the requirements of 3.4. The ingredients may be mixed in whole or in part before dehydrating or may be completely dry blended after dehydration.

* 3.3.3 Pasteurization.- Milk or skimmed milk shall be pasteurized by one of the following methods: (1) heating every particle to at least 145°F. and holding at this temperature for at least 30 minutes, (2) heating every particle to at least 161°F. and holding at this temperature for at least 16 seconds, or (3) using any other time and temperature combinations proved equally efficient by the U.S. Public Health Service (USPHS) and approved by the appropriate state or local authority. If any of the ingredients from 3.1 are added to the basic skimmed milk prior to entry into a condensing unit or if the product is formulated by blending concentrated fluid ingredients the resulting product shall be pasteurized by one of the following methods: (1) heating every particle to at least 155°F. and holding at this temperature for at least 30 minutes or (2) any other time and temperature combination proved equally efficient by the USPHS and approved by the appropriate state or local authority.

* 3.3.4 Dispersibility.- The finished product or one or more of the ingredients prior to dry blending may be partially rehydrated and redried in order to improve the dispersing characteristics of the product.

* 3.3.5 Spray drying.- Spray drying shall be of the continuous product removal type. Wet-collectors shall not be used. The dry product shall be cooled to 100°F. or lower prior to intermediate or final packaging.

3.4 Formulation.- Formulation shall correspond with the compositions specified in 3.4.1 or 3.4.2.

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* 3.4.1 Type I.- The following composition shall constitute one package as described in table VI (sufficient for producing 40 quarts of recombined, filled milk):

Spray process, low heat, nonfat dry milk	8.00 pounds
Lactose	4.00 ounces
Salt	0.30 ounces
Hydrogenated vegetable fat	3.00 pounds

This formulation shall be fortified with 80,000 I.U. of Vitamin A and 16,000 I.U. of Vitamin D.

* 3.4.2 Type II.- The following composition shall constitute one package as described in table VI (sufficient for producing 40 quarts of recombined, filled, chocolate milk):

Spray process, low heat, nonfat dry milk	8.50 pounds
Salt	0.30 ounces
Hydrogenated vegetable fat	3.00 pounds
Cocoa	1.25 pounds
Sugar	4.00 pounds

This formulation shall be fortified with 80,000 I.U. of Vitamin A and 16,000 I.U. of Vitamin D.

3.5 Finished product.-

3.5.1 Physical requirements.- Both types shall be free-flowing powders, free from lumps that do not fall apart under light pressure and free from foreign material. Type I product shall have a white to medium cream color and, when reconstituted by adding 6.4 fluid ounces of 110° ± 5°F. water to 1 ounce by weight of powder, shall have a color and flavor typical of fresh whole milk and shall be free from stale, oxidized, tallowy, rancid or other objectionable flavors and odors. Type II product shall have a light to medium chocolate color and, when reconstituted by adding 4.5 fluid ounces of 110° ± 5°F. water to 1 ounce by weight of powder, shall have a medium chocolate color, a pleasant tasting chocolate-milk flavor, and shall be free from stale, oxidized, tallowy, rancid, or other objectionable flavors and odors.

3.5.2 Analytical requirements.- The finished products shall comply with the following analytical requirements:

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	Type I	Type II
Moisture, not more than	2.5 percent	2.5 percent
Fat, not less than	26.6 percent	17.0 percent
Protein, not less than	26.0 percent	18.0 percent
Lactose, not less than	35.0 percent	-
Sucrose, not less than	-	23.5 percent
Vitamin A, not less than	80,000 I.U. 2/	80,000 I.U. 3/
Vitamin D, not less than	16,000 I.U. 2/	16,000 I.U. 3/
Iron, not more than 1/	15 p.p.m	15 p.p.m
Copper, not more than 1/	1.5 p.p.m	1.5 p.p.m
Bacterial estimate, not more than	30,000 per gram	30,000 per gram
Coliform count, not more than	90 per gram	90 per gram
Salmonellae test	Negative	Negative
Phosphatase	4/	4/

1/ If all equipment and utensils coming into direct contact with the product during processing do not show exposed surfaces of iron or copper and testing of product indicated compliance with requirements, subsequent testing for iron and copper shall be on a reduced basis. Testing one lot from each contract or spot testing on a 3-month basis may be considered adequate, provided no exposed surfaces of iron or copper are observed, otherwise each production lot shall be tested.

2/ Per 15 pounds 5 ounces of product.

3/ Per 17 pounds 1 ounce of product.

4/ Phosphatase acitivity for both types shall be less than 1 microgram of phenol per ml. of product when tested in accordance with the Scharer Rapid Phosphatase Test.

3.6 Age of pack.- The product shall be delivered within 60 days following manufacture.

3.7 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and General Regulations for Its Enforcement.

* 3.8 Plant qualification.- The product shall be prepared, processed and packaged in plants which are approved under the USDA General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service, and listed by the USDA in a document titled 'Dairy Plants Surveyed and Approved for USDA Grading Service' and approved for the finished product described in this specification, or which have a pasteurization plant compliance rating of 90 or

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more as certified by a State Milk Sanitation rating officer and listed in the document titled 'Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers' published quarterly by the U. S. Public Health Service, or in plants meeting the requirements of MIL-STD-671.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Plant qualification conditions.- The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification conditions of 3.8.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

* 4.3.1.1 Inspection of milk and milk products.-

* 4.3.1.1.1 Animal sources.- Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in herds designated by the U.S. Department of Agriculture as tuberculosis-free and from herds designated brucellosis-free or in the process of being so designated. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest office of the Agricultural Research Service of the Animal Health Division. (Location of this office may be ascertained by consulting the publication entitled Directory of Consumer Protection Programs, Establishments, Circuits and Officials.) Nonconformance to either of the above requirements shall be cause for rejection of the lot.

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* 4.3.1.1.2 Milk and milk product component sources.- Examination shall be made to determine that all milk and milk products used in the manufacture of the finished product shall have been processed, packaged and shipped by establishments approved as Military sources of supply as listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement or listed by the U.S. Department of Agriculture in the document entitled 'Dairy Plants Surveyed and Approved for USDA Grading Service' and approved for the component products used to manufacture the finished product described in this specification.

* 4.3.1.1.3 Examination of raw whole milk.- Examination and testing of raw whole milk shall be made to determine compliance with the requirements of 3.1.1, as concerns appearance and odor, bacterial estimate and sediment at the approximate time of use. Records of all required examinations and tests shall be maintained and made available for review. Failure to comply with one or more of the above referenced requirements, reflected by actual examination and tests or by examination of records, shall be cause for rejection of the involved lot of finished product.

* 4.3.1.1.4 Examination of nonfat dry milk and other sources of nonfat milk solids.- Examination of nonfat dry milk, skimmed milk and concentrated skimmed milk shall be made to determine compliance with the requirements of 3.1.1.1, as concerns grade, type, source, identity, age at time of use (as applicable) and condition. The determination for grade, identity, age shall be ascertained by an examination of USDA grade shield, plant records, production dates, labels, invoices or other valid documents. The sample unit for condition examination shall be approximately 6 fluid ounces of the liquid products from each of 5 primary containers or all containers if less than 5 form a lot. Noncompliance with one or more identity or condition requirements or absence of pertinent records to determine compliance with applicable requirements shall be cause for rejection of the involved lot or finished product made therefrom.

* 4.3.1.1.5 Examination of vegetable fat and cocoa.- Examination shall be made in accordance with the applicable examination criteria for product characteristics contained in MIL-F-35081 and JJJ-C-501, except that the sampling plans and AQLs shall not apply. The sample for examination shall be an approximately 8-ounce composite derived from each of five primary containers or all containers if the lot contains less than 5 containers. Nonconformance to one or more requirements shall be cause for rejection of the component lot or finished product made therefrom.

* 4.3.1.1.6 Examination of lactose, salt, vanilla flavoring, stabilizing salts or dispersing agents, emulsifying agents, sugar, vitamins A and D, antioxidants and coloring.- Each ingredient, if used, shall be examined to determine compliance with the requirements of the pertinent ingredient paragraph, as concerns identity (type, FDA approved, components, etc.). Such determinations shall be

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ascertained by examination of pertinent labels, invoices or other valid documents. In addition, each of the ingredients, as applicable, shall be examined organoleptically to determine compliance with the specified condition requirements. The sample unit for examination shall be approximately 8 ounces taken from each of 5 primary containers or all containers if the lot contains less than 5 containers. Noncompliance with one or more identity or condition requirements shall be cause for rejection of the component lot or finished product made therefrom.

* 4.3.2 Testing of components.- Testing of components described in this specification shall be as shown in table I. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE I.- Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to	Test ref and reqmt para
Polyethylene	1/2 sq. yd. or equivalent in polyethylene bags	Yards or bags	S-1	Thickness	Nearest 0.001 inch	5.1
Chipboard pad	1 chipboard pad	Chipboard pads	S-1	Thickness	Nearest 0.001 inch	5.2.1

* 4.3.3 Preprocess and process examinations.- Examination shall be made during preprocessing and processing to determine compliance with the requirements of 3.3.1 (temperature and time intervals for milk product storage), of 3.3.2 (blending), of 3.3.3 (pasteurization, regulatory times and temperatures), of 3.3.4 (optional dispersibility procedures) and of 3.3.5 (spray drying - type of operation and procedures). In addition, examination shall be made to determine compliance with the formulation requirements of 3.4. Records of all applicable operating control data and product formulation shall be maintained. Noncompliance with one or more of the above referenced requirements reflected by actual examination or by examination of records, shall be cause for rejection of the involved quantity of finished product.

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* 4.3.4 Examination of end item.- Classification of defects found during examination shall be in accordance with tables III through V and 4.3.4.1 and 4.3.4.2. The examination criteria for the above tables shall be as shown in table II. Acceptable quality levels (AQL's) shall be expressed as percent defective for table III and in terms of defects per hundred units for tables IV and V.

TABLE II.- End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major A	Major B	Minor	Total
III	S-3	1 filled and sealed primary container	Primary containers	-	-	2.5	-
IV	I	1 filled and sealed primary container	Primary containers	0.25	1.0	-	4.0
V	S-2	Contents of 1 primary container	Primary containers	1/	1.0	-	4.0

1/ See footnote 1/ to table V.

TABLE III.- Examination for net weight 1/ 2/

Category	Defect
Minor	Type I
201	Net weight less than 11 pounds, 2 ounces.
	Type II
202	Net weight less than 16 pounds, 10 ounces.

1/ Lot shall be rejected if sample data indicate lot average net weight is less than the specified net weight.

2/ Report results to nearest one ounce.

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TABLE IV.- External examination of polyethylene bag

Category			Defect
Major A	Major B	Minor	
101			Tear, hole or open seal.
	151		Not type or size (dimensions) specified.
	152		Not sealed as specified.
	153		Not clean.
	154		Excessive air in bag.
	155		Nomenclature missing, incorrect or illegible.
		201	Has objectionable odor.

TABLE V.- Examination of product characteristics 1/

Category			Defect
Major A	Major B	Minor	Dry product
101			Presence of foreign material, e.g. dirt, insect(s), insect part(s), wood, glass, paper, etc.
	151		Presence of lumps which do not fall apart with light finger pressure.
	152		Powder not free-flowing.
		201	Color for specific type not as specified.

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TABLE V.- Examination of product characteristics 1/ (cont'd)

Category			Defect
Major A	Major B	Minor	Reconstituted product (see 3.5.1 for procedure)
102			Presence of objectionable flavors and odors, e.g. stale, oxidized, tallowy, rancid, etc.
	153		Flavor not typical of fresh whole milk (type I), or not a pleasant tasting chocolate milk flavor (type II).
		202	Color not as specified.

1/ Presence of one or more Major A defects shall be cause for rejection of the involved lot of finished product.

* 4.3.4.1 Examination of shipping container.- Examination of the filled and closed shipping container shall be made in accordance with the applicable examination criteria in the appendix to PPP-B-636. In addition, the following defects shall be included in the examination: Major: (1) Markings missing, incorrect or illegible; (2) Directions for use sheet missing or not as specified. For level B pack, the following defects shall be included in the examination: Major: (1) Incorrect number of filled bags in container, as applicable; (2) Missing chipboard pad or pressure sensitive tape, as applicable; Minor: Bags not flat stacked. For level C pack examination shall be in accordance with the aforementioned specification except only the defects for marking and missing chipboard or pressure sensitive tape as applicable shall apply.

* 4.3.4.2 Examination of age of product at time of shipment.- Examination shall be made at time of shipment to determine compliance with the requirements of 3.6, as concerns age of pack. This determination shall be ascertained by examination of records, codes, marked date of pack or other valid means. Noncompliance with the above referenced requirements or missing age markings or records, shall be cause for rejection of the lot.

* 4.4 Sampling procedures and acceptance criteria for testing of finished product.- The finished product shall be tested to determine compliance with the analytical and microbiological requirements of 3.5.2, as applicable. Procedures for testing shall be in accordance with 4.5. Analytical requirements

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shall be applicable to the lot average. The sample unit for microbiological determinations shall be an approximate 1 pound sample taken aseptically from a primary container. Lot size shall be expressed in terms of primary containers and the sample size shall be the number of sample units indicated by inspection level S-2. The AQL, expressed as percent defective for the standard plate count and the coliform count, shall be 1.5. A nonconformance to a microbiological test requirement shall be scored as a defect. The sample unit used for lot average test requirements may be a composite taken from the number of sample units used for the microbiological determinations or shall be an orginal 1-pound composite using the same inspection level. Results shall be reported to the same unit or decimal place as specified for the specific requirement, except for microbiological counts salmonella and vitamin determinations. Results for counts shall be reported in accordance with Standard Methods for the Examination of Dairy Products and for vitamins A and D, to the nearest 50 I.U.'s. Results for salmonella shall be reported as pass (negative) or fail (positive). As applicable, nonconformance to one or more of the above referenced analytical requirements, other than microbiological, shall be cause for rejection of the involved lot of finished product.

4.5 Test methods.- Examination and test procedures which differ from those specified herein unless otherwise excepted may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.5.1 Chemical analyses.- Chemical analyses shall be made in accordance with methods from publications referenced below:

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Test	Source	Method
Finished product:		
Fat	Chapter: Dairy Products, Section: Dried Milk, Nonfat Dry Milk and Malted Milk 1/	Fat in dried milk
Protein	Chapter: Dairy Products Section: Dried Milk Nonfat Dry Milk and Malted Milk 1/	Protein (use 15 grams K_2SO_4 instead of 10 grams.) % N x 6.38
Moisture	ADMI Bulletin No. 916 2/	Toluene Distillation
Sucrose	Chapter: Dairy Products, Section: Sweetened Condensed Milk 1/	Sucrose
Lactose	Chapter: Dairy Products, Section: Milk 1/, 3/	Lactose
Iron	ADMI Bulletin No. 916 2/	Iron
Copper	ADMI Bulletin No. 916 2/	Copper

1/ Official Methods of Analysis of the Association of Official Analytical Chemists.

2/ American Dry Milk Institute Bulletin No. 916, Standards for Grades of Dry Milks including Methods of Analysis.

3/ Dilute 10 grams of product to 100 milliliters and proceed as for milk.

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4.5.1.1 Vitamin A.- Vitamin A content shall be determined by the colorimetric method specified in Methods of Vitamin Assay of the Association of Vitamin Chemists.

* 4.5.1.2 Vitamin D.- Vitamin D content shall be determined by the method described in Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Vitamins and Other Nutrients, Section: Vitamin D.

* 4.5.2 Bacteriological analyses.- Bacteriological analyses shall be made in accordance with one of the following methods from Standard Methods for the Examination of Dairy Products:

Test	Source	Method
Finished product:	Chapter: Microbiological Methods for Concentrated Milk and Dry Milk, Section: Dry Milk	
Bacterial estimate (as standard plate count)		Agar Plate
Coliform		Coliform Group (Coliform Test with Solid Media using Desoxycholate Lactose Agar)

* 4.5.2.1 Salmonella.- Salmonella test procedures shall be conducted as outlined in Bacteriological Analytical Manual of the U.S. Department of Health, Education and Welfare, Food and Drug Administration for Dried Milk.

* 4.5.3 Phosphatase.- Phosphatase shall be determined in accordance with the following method from Standard Methods for the Examination of Dairy Products, Chapter: Phosphatase Methods to Determine Pasteurization, Method: Scharer Rapid Phosphatase Test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C.- The product, of the net weight and bag size specified in table VI, shall be filled into a bag made from polyethylene not less than 0.004-inch thick when tested in accordance with L-P-378. After filling, as much air as possible shall be excluded by pressure on the sides of the bag before final sealing. The bag shall be heat sealed with seals having a minimum width of 3/16 inch. Alternatively, two heat seals, each with a minimum width of 1/16 inch, may be utilized.

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TABLE VI.- Net weight and bag size

Type of product	Net weight		Approximate bag size (inside dimensions)
	Pounds	Ounces	Inches
I	11	4	20 by 13
II	16	12	20 by 13

5.2 Packing.- Packing shall be level B or C as specified (see 6.1).

5.2.1 Level B.- Five filled bags of type I product, or 4 filled bags of type II product, shall be flat stacked into a snug-fitting fiberboard box constructed and closed in accordance with style CSSC, V3c of PPP-B-636. Strapping is not required. When the bottom flaps are stitched, a chipboard pad not less than 0.018-inch thick shall completely cover the bottom of the box. Stitches in the manufacturer's joint shall be covered with a pressure-sensitive tape.

5.2.2 Level C.- The shipping container shall be constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure of PPP-B-636. Bags shall be protected from stitches as specified in 5.2.1.

5.2.2.1 When specified (see 6.2), the shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Labeling and marking.-

5.3.1 Labeling.- Bags shall be labeled with the nomenclature of the product. Directions for use, as shown in the appendix, shall be printed on a sheet of good quality paper approximately 8-1/2 by 11 inches in size. The paper shall take clear, readable impressions without blurring or smudging. No less than 8-point bold-face type shall be used in printing. One sheet shall be placed in each shipping container before closing.

5.3.2 Marking.- The shipping container shall be marked in accordance with MIL-STD-129. In addition, the following marking shall be applied to the shipping container in letters one inch high:

DIRECTION FOR USE SHEET INSIDE

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6. NOTES

6.1 Intended use and product information.- The products covered by this specification are intended primarily for the production of a reconstituted, filled chocolate milk for use by the Air Force. The powdered products are packaged in sealed flexible bags with sufficient dry product in each bag to produce 40 quarts of finished filled milk or chocolate milk when reconstituted and processed according to the instructions supplied by the manufacturer of special equipment specifically adapted for this purpose. Refrigeration for the dry products is not necessary; however, stability is improved if the products are stored in a cool location.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of product required (see 1.2).
(c) Level of packing required (see 5.2).
(d) When packing specified in 5.2.2.1 is required.

6.3 Packing specified in 5.2.2.1 is intended for direct shipment from the supply source to the consumer for immediate use within CONUS.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections and deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP-SS

Preparing activity:

Army - GL

Project No. 8910-0168

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APPENDIX

10. Instructions for blending ingredients in mechanical cow (or equal) dispenser to produce type I or II product.

1. Add potable water (36-1/4 quarts) to tank, using special measuring rod.
2. Turn on tank agitator and steam valve.
3. When water in tank reaches 110°F., add complete contents of this package.
4. Continue heating until thermometer reads 162°F.
5. Turn off steam valve, leaving agitator on.
6. Open oiler delivery valve on homogenizer.
7. Turn on compressor.
8. When cooler suction line is cold, turn on homogenizer.
9. When homogenizer is at full speed, open tank valve fully, allowing milk to pass through homogenizer and over surface cooler.
10. Runoff temperature should be 40° to 50°F. At beginning of cycle should temperature exceed 50°F., throttle back tank valve slightly to compensate for hydrostatic head pressure.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MILK, FILLED, DRY, PLAIN OR CHOCOLATE, FORTIFIED MIL-M-43241A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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